

Math 1512 Calculus 1 Review: Selected Solutions

Fall 2026

Problem numbers match the review sheet. Each problem is restated before its solution.

1. Memorization.

- (a) State the definition of f is continuous at $x = a$.

Solution. f is continuous at $x = a$ if $\lim_{x \rightarrow a} f(x) = f(a)$.

This one equation hides three requirements: $f(a)$ is defined, $\lim_{x \rightarrow a} f(x)$ exists, and the two are equal. A hole fails the third, a jump fails the second, a vertical asymptote fails all three. ■

- (b) State both limit definitions of $f'(a)$, the derivative of f at $x = a$.

Solution.

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}.$$

Both are limits of slopes of secant lines through $(a, f(a))$. The second form just names the horizontal gap $h = x - a$. ■

- (l) State both parts of the Fundamental Theorem of Calculus.

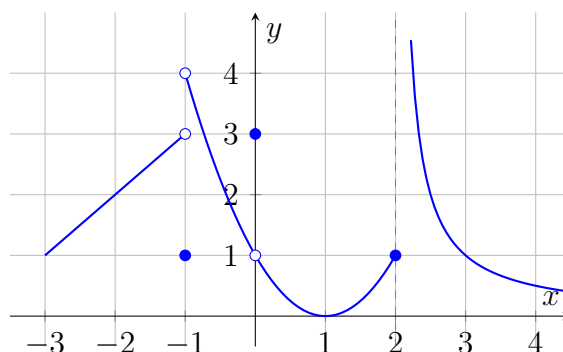
Solution. Let f be continuous on $[a, b]$.

Part 1. If $g(x) = \int_a^x f(t) dt$, then $g'(x) = f(x)$. The rate the area grows is the height of the graph at the moving edge.

Part 2. If F is any antiderivative of f , meaning $F' = f$, then $\int_a^b f(x) dx = F(b) - F(a)$. Adding up tiny changes in F gives the total change in F . ■

2. Limits from a graph.

- (a-c) Let g be the function graphed below (a filled dot is a point on the graph, an open dot is not). Find $g(-1)$, $\lim_{x \rightarrow -1^-} g(x)$, $\lim_{x \rightarrow -1^+} g(x)$, $\lim_{x \rightarrow -1} g(x)$, then $g(0)$ and $\lim_{x \rightarrow 0} g(x)$, then $\lim_{x \rightarrow 2^-} g(x)$, $\lim_{x \rightarrow 2^+} g(x)$, $\lim_{x \rightarrow 2} g(x)$.



Solution. At $x = -1$ the filled dot gives $g(-1) = 1$. From the left the line climbs to the open dot at height 3, from the right the parabola comes down from the open dot at height 4.

$$\lim_{x \rightarrow -1^-} g(x) = 3, \quad \lim_{x \rightarrow -1^+} g(x) = 4,$$

so $\lim_{x \rightarrow -1} g(x)$ does not exist, since the one-sided limits disagree.

At $x = 0$ the dot sits at $g(0) = 3$, but the curve on both sides heads to the open dot at height 1. So $\lim_{x \rightarrow 0} g(x) = 1$. The value of g at 0 has no effect on the limit.

At $x = 2$ the parabola arrives at the filled dot, so $\lim_{x \rightarrow 2^-} g(x) = 1$. To the right the graph shoots up along the dashed asymptote, so $\lim_{x \rightarrow 2^+} g(x) = \infty$. The two-sided limit does not exist. ■

- (d-e) At which x values in $(-3, 4)$ is g not continuous? Classify each discontinuity. At which of these is g continuous from the left, or from the right?

Solution. g is not continuous at $x = -1, 0, 2$.

At $x = -1$ the one-sided limits are 3 and 4: a jump. At $x = 0$ the limit exists ($= 1$) but is not $g(0) = 3$: removable, since redefining one point fixes it. At $x = 2$ the right-hand limit is infinite: an infinite discontinuity.

Continuous from the left at a means $\lim_{x \rightarrow a^-} g(x) = g(a)$. This holds only at $x = 2$, where both equal 1. No point is continuous from the right: at -1 we compare 4 with 1, at 0 we compare 1 with 3, and at 2 the right limit is infinite. ■

- (f) Suppose $\lim_{x \rightarrow a} f(x) = 4$ and $\lim_{x \rightarrow a} h(x) = -2$. Evaluate $\lim_{x \rightarrow a} [3f(x) - 5h(x)]$, $\lim_{x \rightarrow a} f(x)h(x)$, $\lim_{x \rightarrow a} \frac{f(x)}{h(x)}$, and $\lim_{x \rightarrow a} \left(\sqrt{f(x)} + h(x)^2 \right)$.

Solution. The limit laws say limits pass through sums, constant multiples, products, quotients (with nonzero bottom limit), powers and roots. So we can replace $f(x)$ by 4 and $h(x)$ by -2 .

$$3(4) - 5(-2) = 22, \quad 4(-2) = -8, \quad \frac{4}{-2} = -2, \quad \sqrt{4} + (-2)^2 = 6.$$

3. Computing limits.

- (a) Evaluate $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x^2 - 4}$.

Solution. Plugging in gives $\frac{0}{0}$, so $x - 2$ divides both.

$$\lim_{x \rightarrow 2} \frac{(x+3)(x-2)}{(x+2)(x-2)} = \lim_{x \rightarrow 2} \frac{x+3}{x+2} = \frac{5}{4}.$$

Canceling is allowed because the limit never uses $x = 2$ itself. ■

- (c) Evaluate $\lim_{x \rightarrow 4} \frac{\sqrt{x}-2}{x-4}$.

Solution. $\frac{0}{0}$ again. Multiply by the conjugate $\sqrt{x} + 2$, or notice $x - 4 = (\sqrt{x} - 2)(\sqrt{x} + 2)$.

$$\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{(\sqrt{x} - 2)(\sqrt{x} + 2)} = \lim_{x \rightarrow 4} \frac{1}{\sqrt{x} + 2} = \frac{1}{4}.$$

■

(f) Evaluate $\lim_{r \rightarrow 0} \frac{\sqrt{4+r} - \sqrt{4-r}}{r}$.

Solution. Rationalize the numerator with the conjugate $\sqrt{4+r} + \sqrt{4-r}$.

$$\frac{(4+r) - (4-r)}{r(\sqrt{4+r} + \sqrt{4-r})} = \frac{2}{\sqrt{4+r} + \sqrt{4-r}} \rightarrow \frac{2}{2+2} = \frac{1}{2}.$$

■

(i) Evaluate $\lim_{x \rightarrow -2} \frac{6}{x^2 - 4}$, considering the one-sided limits.

Solution. The top is 6 and the bottom $(x-2)(x+2)$ goes to 0, so each one-sided limit is $\pm\infty$. Track signs.

For x slightly less than -2 : $x+2 < 0$ and $x-2 < 0$, so the bottom is a small positive number and $\lim_{x \rightarrow -2^-} = \infty$. For x slightly more than -2 : $x+2 > 0$ and $x-2 < 0$, so the bottom is a small negative number and $\lim_{x \rightarrow -2^+} = -\infty$.

The one-sided limits disagree, so the limit does not exist.

■

(j) Evaluate $\lim_{t \rightarrow 3} \frac{t^3}{(t-1)(t-3)^2}$.

Solution. Near $t = 3$ the top is near 27 and $t-1$ is near 2. The factor $(t-3)^2$ is a small positive number on both sides of 3, because it is a square.

So the quotient is a large positive number on both sides and $\lim_{t \rightarrow 3} \frac{t^3}{(t-1)(t-3)^2} = \infty$.

■

4. Squeeze Theorem.

(a) Evaluate $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{3}{x}\right)$.

Solution. $\sin\left(\frac{3}{x}\right)$ oscillates and has no limit, so the product law does not apply (this answers (d)). But it is bounded: $-1 \leq \sin\left(\frac{3}{x}\right) \leq 1$. Multiplying by $x^2 \geq 0$,

$$-x^2 \leq x^2 \sin\left(\frac{3}{x}\right) \leq x^2.$$

Both bounds go to 0, so by the Squeeze Theorem the limit is 0. The x^2 crushes the oscillation.

■

(c) If $2x - 1 \leq f(x) \leq x^2 - 2x + 3$ for all x , find $\lim_{x \rightarrow 2} f(x)$.

Solution. $\lim_{x \rightarrow 2} (2x - 1) = 3$ and $\lim_{x \rightarrow 2} (x^2 - 2x + 3) = 3$. Since f is squeezed between them, $\lim_{x \rightarrow 2} f(x) = 3$.

The bounds touch at $x = 2$ because $(x^2 - 2x + 3) - (2x - 1) = (x - 2)^2$.

■

5. Limits at infinity.

- (d) Evaluate $\lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2+1}}{x+3}$.

Solution. Pull x^2 out of the root: $\sqrt{4x^2+1} = \sqrt{x^2} \sqrt{4 + \frac{1}{x^2}} = |x| \sqrt{4 + \frac{1}{x^2}}$. For $x < 0$, $|x| = -x$.

$$\frac{-x \sqrt{4 + \frac{1}{x^2}}}{x \left(1 + \frac{3}{x}\right)} = \frac{-\sqrt{4 + \frac{1}{x^2}}}{1 + \frac{3}{x}} \rightarrow -2.$$

The sign makes sense: the top is positive and the bottom is negative for very negative x . ■

- (e) Evaluate $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 6x} - x)$.

Solution. This is $\infty - \infty$, which is not determined yet. Multiply by the conjugate over itself.

$$\frac{(x^2 + 6x) - x^2}{\sqrt{x^2 + 6x} + x} = \frac{6x}{x \left(\sqrt{1 + \frac{6}{x}} + 1\right)} = \frac{6}{\sqrt{1 + \frac{6}{x}} + 1} \rightarrow \frac{6}{2} = 3.$$

■

- (g) Find all vertical and horizontal asymptotes of $f(x) = \frac{x^2-x-2}{x^2-4}$. Why is $x = 2$ not a vertical asymptote?

Solution. Factor: $f(x) = \frac{(x-2)(x+1)}{(x-2)(x+2)} = \frac{x+1}{x+2}$ for $x \neq 2$.

At $x = -2$ the top is $-1 \neq 0$ and the bottom is 0, so $x = -2$ is a vertical asymptote.

At $x = 2$ the bad factor cancels and $\lim_{x \rightarrow 2} f(x) = \frac{3}{4}$, a finite number. So the graph has a hole at $(2, \frac{3}{4})$, not an asymptote.

The leading terms are both x^2 , so $\lim_{x \rightarrow \pm\infty} f(x) = 1$ and $y = 1$ is the horizontal asymptote. ■

6. Continuity and the IVT.

- (d) Let $f(x) = \begin{cases} x+2 & x < -1 \\ x^2 & -1 \leq x \leq 1 \\ 3-x & x > 1 \end{cases}$. Is f continuous at $x = -1$ and at $x = 1$? Where is f not differentiable?

Solution. At $x = -1$: $f(-1) = 1$, $\lim_{x \rightarrow -1^-} (x+2) = 1$, $\lim_{x \rightarrow -1^+} x^2 = 1$. All agree, so f is continuous at -1 .

At $x = 1$: $f(1) = 1$ and $\lim_{x \rightarrow 1^-} x^2 = 1$, but $\lim_{x \rightarrow 1^+} (3-x) = 2$. The limit does not exist, so f is not continuous at 1 (a jump).

f is not differentiable at $x = 1$, since a differentiable function is continuous. It is also not differentiable at $x = -1$: the graph is continuous there but has a corner, with slope 1 from the left and slope $2x = -2$ from the right. ■

- (f) Find c so that $f(x) = \begin{cases} \frac{x^2}{x+2} & x \leq 2 \\ 3x + c & x > 2 \end{cases}$ is continuous at $x = 2$.

Solution. $f(2) = \frac{4}{4} = 1$, and this is also the left limit. The right limit is $\lim_{x \rightarrow 2^+} (3x + c) = 6 + c$. Continuity needs $6 + c = 1$, so $c = -5$. ■

- (g) Find a and b so that $f(x) = \begin{cases} 2 & x < 1 \\ ax + b & 1 \leq x < 4 \\ 8 & x \geq 4 \end{cases}$ is continuous everywhere.

Solution. Each piece is continuous, so only the seams at $x = 1$ and $x = 4$ matter. The middle line must start at height 2 and end at height 8.

$$a + b = 2, \quad 4a + b = 8.$$

Subtracting, $3a = 6$, so $a = 2$ and $b = 0$. The middle piece is $2x$. ■

- (i) Show that $x^3 + x - 1 = 0$ has a solution in $(0, 1)$.

Solution. Let $f(x) = x^3 + x - 1$. It is a polynomial, so it is continuous on $[0, 1]$. $f(0) = -1 < 0$ and $f(1) = 1 > 0$.

Since 0 is between $f(0)$ and $f(1)$, the Intermediate Value Theorem gives a number c in $(0, 1)$ with $f(c) = 0$. A continuous graph cannot get from below the axis to above it without crossing. ■

7. The derivative as a limit.

- (a) Find the equation of the line tangent to $f(x) = \sqrt{x}$ at $x = 4$, as a limit of slopes of secant lines.

Solution. The secant line through $(4, 2)$ and $(x, \sqrt{x})$ has slope $\frac{\sqrt{x}-2}{x-4}$. As $x \rightarrow 4$ the second point slides onto the first and the secant turns into the tangent.

$$m = \lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4} = \lim_{x \rightarrow 4} \frac{1}{\sqrt{x} + 2} = \frac{1}{4}.$$

The tangent line is $y - 2 = \frac{1}{4}(x - 4)$, that is $y = \frac{x}{4} + 1$. ■

- (c) Use the limit definition of the derivative to find $f'(x)$ if $f(x) = \frac{2}{x^2}$.

Solution. Combine the fractions in the numerator first.

$$\frac{f(x+h) - f(x)}{h} = \frac{1}{h} \left(\frac{2}{(x+h)^2} - \frac{2}{x^2} \right) = \frac{2(x^2 - (x+h)^2)}{h x^2 (x+h)^2} = \frac{-2(2x+h)}{x^2(x+h)^2}.$$

Now h is gone from the bottom, so let $h \rightarrow 0$: $f'(x) = \frac{-4x}{x^4} = -\frac{4}{x^3}$. This matches the power rule applied to $2x^{-2}$. ■

- (f) A ball's height is $h(t) = 200 - 16t^2$ feet after t seconds. Find the slope of the secant line between $t = 1$ and $t = 3$, then the instantaneous velocity at $t = 2$ as a limit.

Solution. $h(1) = 184$ and $h(3) = 56$, so the average velocity is $\frac{56-184}{3-1} = -64$ ft/s.

For the instantaneous velocity at $t = 2$ we use a small time step k (not h , which already names the height).

$$\lim_{k \rightarrow 0} \frac{h(2+k) - h(2)}{k} = \lim_{k \rightarrow 0} \frac{-16(4k + k^2)}{k} = \lim_{k \rightarrow 0} (-64 - 16k) = -64 \text{ ft/s.}$$

The two agree. For a parabola the secant slope over an interval equals the tangent slope at the midpoint. ■

- (h) Each limit is the derivative of some f at some a . Find f and a , then evaluate:

$$\lim_{h \rightarrow 0} \frac{(1+h)^3 - 1}{h} \text{ and } \lim_{x \rightarrow 3} \frac{\frac{1}{x} - \frac{1}{3}}{x-3}.$$

Solution. The first has the shape $\frac{f(a+h)-f(a)}{h}$ with $f(x) = x^3$, $a = 1$. So it equals $f'(1) = 3(1)^2 = 3$.

The second has the shape $\frac{f(x)-f(a)}{x-a}$ with $f(x) = \frac{1}{x}$, $a = 3$. So it equals $f'(3) = -\frac{1}{3^2} = -\frac{1}{9}$. Each is the slope of the tangent line to $y = f(x)$ at $x = a$. ■

8. Differentiation rules.

- (c) Find y' if $y = \frac{x^2+1}{2x+1}$.

Solution. Quotient rule: bottom times derivative of top, minus top times derivative of bottom, over bottom squared.

$$y' = \frac{2x(2x+1) - (x^2+1)(2)}{(2x+1)^2} = \frac{2x^2+2x-2}{(2x+1)^2}.$$

■

- (f) At what points (x, y) is the tangent line to $g(x) = \frac{x}{x^2+4}$ horizontal?

Solution. A horizontal tangent means $g'(x) = 0$.

$$g'(x) = \frac{(x^2+4) - x(2x)}{(x^2+4)^2} = \frac{4-x^2}{(x^2+4)^2}.$$

The bottom is never 0, so $g'(x) = 0$ exactly when $x = \pm 2$. The points are $(2, \frac{1}{4})$ and $(-2, -\frac{1}{4})$. ■

- (h) A particle has position $s(t) = t^3 - 9t^2 + 24t$ meters at time t seconds. Find $v(t)$ and $a(t)$, with units. When is the particle at rest?

Solution.

$$v(t) = s'(t) = 3t^2 - 18t + 24 \text{ m/s}, \quad a(t) = v'(t) = 6t - 18 \text{ m/s}^2.$$

At rest means $v = 0$: $3(t-2)(t-4) = 0$, so $t = 2$ s and $t = 4$ s. Between these times $v < 0$ and the particle moves backward. ■

9. Trig derivatives.

- (b) Evaluate $\lim_{x \rightarrow 0} \frac{\sin(4x)}{3x}$, $\lim_{x \rightarrow 0} \frac{\tan(2x)}{x}$, and $\lim_{x \rightarrow 0} \frac{\sin(3x)}{\sin(5x)}$.

Solution.

$$\frac{\sin(4x)}{3x} = \frac{4}{3} \cdot \frac{\sin(4x)}{4x} \longrightarrow \frac{4}{3}.$$

$$\frac{\tan(2x)}{x} = 2 \cdot \frac{\sin(2x)}{2x} \cdot \frac{1}{\cos(2x)} \longrightarrow 2 \cdot 1 \cdot 1 = 2.$$

$$\frac{\sin(3x)}{\sin(5x)} = \frac{3x}{5x} \cdot \frac{\sin(3x)/(3x)}{\sin(5x)/(5x)} \longrightarrow \frac{3}{5}.$$

■

- (e) Find y' if $y = \frac{\sec x}{1 + \tan x}$.

Solution.

$$y' = \frac{\sec x \tan x (1 + \tan x) - \sec x \sec^2 x}{(1 + \tan x)^2} = \frac{\sec x (\tan x + \tan^2 x - \sec^2 x)}{(1 + \tan x)^2}.$$

Since $\sec^2 x = 1 + \tan^2 x$, the parentheses simplify to $\tan x - 1$, and $y' = \frac{\sec x (\tan x - 1)}{(1 + \tan x)^2}$. ■

- (g) Find all points (x, y) with $0 \leq x < 2\pi$ where $y = x + 2 \cos x$ has a horizontal tangent line.

Solution. $y' = 1 - 2 \sin x = 0$ gives $\sin x = \frac{1}{2}$, so $x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$.

The points are $(\frac{\pi}{6}, \frac{\pi}{6} + \sqrt{3})$ and $(\frac{5\pi}{6}, \frac{5\pi}{6} - \sqrt{3})$. ■

10. Chain rule.

- (c) Find the equation of the line tangent to $x(t) = \sin(\sin(\sin(t)))$ when $t = \pi$.

Solution. Differentiate from the outside in, one layer at a time.

$$x'(t) = \cos(\sin(\sin t)) \cdot \cos(\sin t) \cdot \cos t.$$

At $t = \pi$, $\sin \pi = 0$, so $x'(\pi) = \cos 0 \cdot \cos 0 \cdot \cos \pi = -1$ and $x(\pi) = 0$. The tangent line is $x = -(t - \pi) = \pi - t$. ■

- (d) Find the equation of the line tangent to $h(t) = (1 + 2t^2)^4$ when $t = 0$.

Solution. $h'(t) = 4(1 + 2t^2)^3 \cdot 4t$, so $h'(0) = 0$ and $h(0) = 1$. The tangent line is horizontal: $h = 1$. ■

- (i) Find all points (x, y) with $0 \leq x < 2\pi$ where $y = 2 \sin x + \cos(2x)$ has a horizontal tangent line.

Solution. The chain rule gives $\frac{d}{dx} \cos(2x) = -2 \sin(2x)$. Then use $\sin(2x) = 2 \sin x \cos x$ to factor.

$$y' = 2 \cos x - 2 \sin(2x) = 2 \cos x - 4 \sin x \cos x = 2 \cos x (1 - 2 \sin x).$$

$\cos x = 0$ gives $x = \frac{\pi}{2}, \frac{3\pi}{2}$, and $\sin x = \frac{1}{2}$ gives $x = \frac{\pi}{6}, \frac{5\pi}{6}$. The points are $(\frac{\pi}{6}, \frac{3}{2})$, $(\frac{\pi}{2}, 1)$, $(\frac{5\pi}{6}, \frac{3}{2})$, $(\frac{3\pi}{2}, -3)$. ■

11. Implicit differentiation.

- (b) Find the equation of the tangent line at $(\pi/8, \pi/8)$ for the curve $\tan(x + y) + \sec(x - y) = 2$.

Solution.

$$\sec^2(x + y) (1 + y') + \sec(x - y) \tan(x - y) (1 - y') = 0.$$

At the point, $x + y = \frac{\pi}{4}$ so $\sec^2(x + y) = 2$, and $x - y = 0$ so $\tan(x - y) = 0$. The equation becomes $2(1 + y') = 0$, so $y' = -1$.

The tangent line is $y - \frac{\pi}{8} = -(x - \frac{\pi}{8})$, that is $y = -x + \frac{\pi}{4}$. ■

- (c) Find the tangent line to $x^3 + y^3 = 9xy$ at $(2, 4)$.

Solution. Check the point: $8 + 64 = 72 = 9(2)(4)$. Differentiate, using the product rule on $9xy$.

$$3x^2 + 3y^2 y' = 9y + 9xy' \implies y' = \frac{3y - x^2}{y^2 - 3x}.$$

At $(2, 4)$, $y' = \frac{12-4}{16-6} = \frac{4}{5}$, and the tangent line is $y = 4 + \frac{4}{5}(x - 2)$. ■

- (e) For $y^2 = x^3 - 3x + 2$, find $\frac{dy}{dx}$, the tangent line at $(2, 2)$, and all points with a horizontal tangent line.

Solution. $2yy' = 3x^2 - 3$, so $y' = \frac{3x^2-3}{2y}$. At $(2, 2)$, $y' = \frac{9}{4}$ and the tangent line is $y = 2 + \frac{9}{4}(x - 2)$.

A horizontal tangent needs the top to be 0 and the bottom not 0: $x = \pm 1$ with $y \neq 0$. At $x = 1$, $y^2 = 0$, so that point is excluded (the formula gives $\frac{0}{0}$ there). At $x = -1$, $y^2 = 4$, giving $(-1, 2)$ and $(-1, -2)$. ■

12. Related rates.

- (a) If x, y, z are functions of t and $x(1 - y) + 5z^3 = y^2 z^2 + x^2 - 3$, find z' when $x = 4$, $y = -2$, $z = 1$, $x' = 9$, $y' = -3$.

Solution. Differentiate every term with respect to t .

$$x'(1 - y) - xy' + 15z^2 z' = 2yy' z^2 + 2y^2 z z' + 2xx'.$$

Plug in: $27 + 12 + 15z' = 12 + 8z' + 72$, so $7z' = 45$ and $z' = \frac{45}{7}$. ■

- (b) A 13 ft ladder leans against a wall. The bottom slides away at 2 ft/s. How fast is the top sliding down when the bottom is 5 ft from the wall?

Solution. Let x be the distance from the wall to the bottom and y the height of the top. Then $x^2 + y^2 = 169$ at every instant, so $2xx' + 2yy' = 0$.

When $x = 5$, $y = 12$. So $y' = -\frac{xx'}{y} = -\frac{5 \cdot 2}{12} = -\frac{5}{6}$ ft/s. The top slides down at $\frac{5}{6}$ ft/s. ■

- (c) A tank is an inverted cone with radius 3 m at the top and height 6 m. Water is pumped in at 4 m³/min. How fast is the water level rising when the water is 4 m deep?

Solution. Let h be the depth and r the radius of the water surface. Similar triangles give $\frac{r}{h} = \frac{3}{6}$, so $r = \frac{h}{2}$ and

$$V = \frac{1}{3}\pi r^2 h = \frac{\pi h^3}{12}, \quad \frac{dV}{dt} = \frac{\pi h^2}{4} \frac{dh}{dt}.$$

At $h = 4$: $4 = 4\pi \frac{dh}{dt}$, so $\frac{dh}{dt} = \frac{1}{\pi}$ m/min. The factor $\frac{\pi h^2}{4}$ is the area of the water surface, which is why the level rises more slowly as the tank fills. ■

- (f) A marble moves along $x^2 + xy + y^2 = 9$. At $(3, 0)$, x is decreasing at 2 ft/s. Is y increasing or decreasing, and how fast?

Solution.

$$2xx' + x'y + xy' + 2yy' = 0.$$

At $(3, 0)$ with $x' = -2$: $2(3)(-2) + 0 + 3y' + 0 = 0$, so $y' = 4$. The y -coordinate is increasing at 4 ft/s. ■

13. Linear approximation.

- (b) If $f(1) = 1$ and $f'(1) = \frac{1}{2}$, what do you expect $f(\frac{3}{2})$ to be? Verify with the linearization.

Solution. Moving right by $\frac{1}{2}$ at slope $\frac{1}{2}$ raises the height by about $\frac{1}{4}$, so $f(\frac{3}{2}) \approx \frac{5}{4}$. The linearization is $L(x) = 1 + \frac{1}{2}(x - 1)$, and $L(\frac{3}{2}) = 1 + \frac{1}{4} = \frac{5}{4}$. ■

- (c) Find the linearization of $f(x) = \sqrt{x+5}$ at $a = 4$. Use it to approximate $\sqrt{9.1}$.

Solution. $f(4) = 3$ and $f'(x) = \frac{1}{2\sqrt{x+5}}$, so $f'(4) = \frac{1}{6}$. Then $L(x) = 3 + \frac{1}{6}(x - 4)$. $\sqrt{9.1} = f(4.1) \approx L(4.1) = 3 + \frac{0.1}{6} = \frac{181}{60} \approx 3.0167$. The true value is 3.01662... ■

- (d) Find the linearization of $f(x) = \sqrt[3]{x}$ at $a = 27$. Use it to approximate $\sqrt[3]{27.27}$.

Solution. $f(27) = 3$ and $f'(x) = \frac{1}{3}x^{-2/3}$, so $f'(27) = \frac{1}{27}$. Then $L(x) = 3 + \frac{x-27}{27}$ and $L(27.27) = 3 + \frac{0.27}{27} = 3.01$. ■

- (f) The radius of a sphere is measured as 10 cm with a possible error of 0.05 cm. Estimate the maximum error in the computed volume.

Solution. $V = \frac{4}{3}\pi r^3$, so $dV = 4\pi r^2 dr = 4\pi(100)(0.05) = 20\pi \approx 62.8$ cm³.

Here $4\pi r^2$ is the surface area: an error dr in the radius adds a thin shell of that area and thickness dr . ■

14. Extreme values.

- (e) Find the absolute max and min of $f(x) = x^4 - 8x^2 + 3$ on $[-1, 3]$.

Solution. $f'(x) = 4x^3 - 16x = 4x(x-2)(x+2)$. The critical numbers in $[-1, 3]$ are 0 and 2 (we drop -2).

$$f(-1) = -4, \quad f(0) = 3, \quad f(2) = -13, \quad f(3) = 12.$$

The absolute max is 12 at $x = 3$ and the absolute min is -13 at $x = 2$. ■

- (g) Find the absolute max and min of $f(x) = \sqrt{9 - x^2}$ on $[-1, 3]$.

Solution. $f'(x) = \frac{-x}{\sqrt{9-x^2}}$ is 0 at $x = 0$ and undefined at $x = 3$, an endpoint anyway.

$f(-1) = 2\sqrt{2}$, $f(0) = 3$, $f(3) = 0$. Max 3 at $x = 0$, min 0 at $x = 3$. The graph is the top of the circle of radius 3, so this matches the picture. ■

- (h) Find the absolute max and min of $f(x) = 2\cos x + \sin(2x)$ on $[0, \pi]$.

Solution. Use $\cos(2x) = 1 - 2\sin^2 x$ to write f' in terms of $\sin x$ only.

$$f'(x) = -2\sin x + 2\cos(2x) = -2(2\sin^2 x + \sin x - 1) = -2(2\sin x - 1)(\sin x + 1).$$

On $[0, \pi]$, $\sin x = -1$ never happens, and $\sin x = \frac{1}{2}$ at $x = \frac{\pi}{6}, \frac{5\pi}{6}$.

$$f(0) = 2, \quad f\left(\frac{\pi}{6}\right) = \frac{3\sqrt{3}}{2}, \quad f\left(\frac{5\pi}{6}\right) = -\frac{3\sqrt{3}}{2}, \quad f(\pi) = -2.$$

Max $\frac{3\sqrt{3}}{2} \approx 2.60$ at $x = \frac{\pi}{6}$, min $-\frac{3\sqrt{3}}{2}$ at $x = \frac{5\pi}{6}$. ■

15. Mean Value Theorem.

- (a) Find every c guaranteed by the MVT for $f(x) = x^2 - 4x + 1$ on $[1, 5]$.

Solution. $f(1) = -2$ and $f(5) = 6$, so the average slope is $\frac{8}{4} = 2$. Solve $f'(c) = 2c - 4 = 2$: $c = 3$. ■

- (c) If $f(2) = -1$ and $f'(x) \leq 3$ for all x , how large can $f(6)$ be?

Solution. By the MVT on $[2, 6]$, $f(6) - f(2) = f'(c)(6 - 2)$ for some c . Since $f'(c) \leq 3$, $f(6) \leq -1 + 3 \cdot 4 = 11$.

The bound is reached by the line $f(x) = -1 + 3(x - 2)$, so 11 is the best possible. ■

- (e) Show that $x^3 + 5x - 2 = 0$ has exactly one real solution.

Solution. Let $f(x) = x^3 + 5x - 2$. At least one: $f(0) = -2 < 0 < 4 = f(1)$, so the IVT gives a root in $(0, 1)$.

At most one: if $f(a) = f(b) = 0$ with $a < b$, the MVT gives c with $f'(c) = \frac{0-0}{b-a} = 0$. But $f'(x) = 3x^2 + 5 \geq 5$. So a second root is impossible. ■

16. Shape of a graph.

- (a) For $f(x) = x^3 - 6x^2 + 9x + 1$, find the intervals of increase and decrease, local extrema, concavity, and inflection points.

Solution. $f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$. So $f' > 0$ on $(-\infty, 1)$ and $(3, \infty)$ (increasing) and $f' < 0$ on $(1, 3)$ (decreasing).

f' changes from $+$ to $-$ at 1: local max $f(1) = 5$. From $-$ to $+$ at 3: local min $f(3) = 1$.
 $f''(x) = 6x - 12$: concave down on $(-\infty, 2)$, concave up on $(2, \infty)$, inflection point $(2, 3)$. The inflection point sits halfway between the max and min, as for every cubic. ■

- (b) Do the same for $f(x) = 3x^4 - 4x^3$.

Solution. $f'(x) = 12x^3 - 12x^2 = 12x^2(x-1)$. The factor $12x^2 \geq 0$, so the sign of f' is the sign of $x-1$: decreasing on $(-\infty, 1)$, increasing on $(1, \infty)$.

Local min $f(1) = -1$. The critical number $x = 0$ is not an extremum, since f' does not change sign there. The tangent is flat but the graph keeps falling.

$f''(x) = 36x^2 - 24x = 12x(3x-2)$: concave up on $(-\infty, 0)$ and $(\frac{2}{3}, \infty)$, concave down on $(0, \frac{2}{3})$. Inflection points $(0, 0)$ and $(\frac{2}{3}, -\frac{16}{27})$. ■

- (f) Let $a > 0$, $c < 0$, and $f(x) = x(ax^2 + bx + c)$ have its inflection point at the origin. Simplify f , then find the roots of f and of f' .

Solution. $f(x) = ax^3 + bx^2 + cx$, so $f''(x) = 6ax + 2b$ and $f''(0) = 2b$. The inflection point at 0 forces $b = 0$, and $f(x) = x(ax^2 + c)$.

Roots of f : $x = 0$, and the quadratic formula on $ax^2 + 0x + c$ gives
 $x = \frac{\pm\sqrt{-4ac}}{2a} = \pm\sqrt{-\frac{c}{a}}$. These are real since $-\frac{c}{a} > 0$.

Roots of $f'(x) = 3ax^2 + c$: $x = \pm\sqrt{-\frac{c}{3a}}$. Since $a > 0$ the graph rises on the far right, so the local max is at $-\sqrt{-\frac{c}{3a}}$ and the local min at $+\sqrt{-\frac{c}{3a}}$. ■

17. Curve sketching.

- (a) Sketch $f(x) = \frac{1}{x^2+3}$, labeling intercepts, asymptotes, extrema, and inflection points.

Solution. Domain $\mathbb{R}$, since $x^2 + 3 > 0$. No x -intercept, y -intercept $(0, \frac{1}{3})$. No vertical asymptote, and $f \rightarrow 0$ as $x \rightarrow \pm\infty$, so $y = 0$ is the horizontal asymptote. f is even.

$$f'(x) = \frac{-2x}{(x^2+3)^2}, \quad f''(x) = \frac{6(x^2-1)}{(x^2+3)^3}.$$

Increasing on $(-\infty, 0)$, decreasing on $(0, \infty)$, local max $(0, \frac{1}{3})$. Concave up on $(-\infty, -1)$ and $(1, \infty)$, concave down on $(-1, 1)$, inflection points $(\pm 1, \frac{1}{4})$. The graph is a bump that flattens toward the x -axis. ■

- (b) Sketch $f(x) = \frac{x^2-1}{x^2-4}$, labeling all key features.

Solution. Domain $x \neq \pm 2$. Intercepts $(\pm 1, 0)$ and $(0, \frac{1}{4})$. Vertical asymptotes $x = \pm 2$ (the top is $3 \neq 0$ there), horizontal asymptote $y = 1$. f is even.

$$f'(x) = \frac{-6x}{(x^2 - 4)^2}, \quad f''(x) = \frac{6(3x^2 + 4)}{(x^2 - 4)^3}.$$

Increasing on $(-\infty, -2)$ and $(-2, 0)$, decreasing on $(0, 2)$ and $(2, \infty)$, local max $(0, \frac{1}{4})$. $3x^2 + 4 > 0$, so f'' has the sign of $x^2 - 4$: concave up for $|x| > 2$, concave down on $(-2, 2)$. No inflection points, since concavity only changes at the asymptotes.

Near the asymptotes: $f \rightarrow -\infty$ as $x \rightarrow 2^-$ and $f \rightarrow \infty$ as $x \rightarrow 2^+$, and symmetrically at -2 . The middle piece is an upside-down bowl through $(\pm 1, 0)$ with top $(0, \frac{1}{4})$, and the outer pieces come down from ∞ toward $y = 1$. ■

18. Optimization.

- (b) A farmer has 2000 feet of fencing to build a rectangular field, fenced on 3 sides, along a straight river. What dimensions maximize the area?

Solution. Let x be each side perpendicular to the river and y the side parallel to it. Then $2x + y = 2000$ and

$$A(x) = x(2000 - 2x), \quad 0 \leq x \leq 1000.$$

$A'(x) = 2000 - 4x = 0$ at $x = 500$, and $A'' = -4 < 0$, so this is the max. The field is 500 ft by 1000 ft with area 500,000 ft². ■

- (d) A box with a square base and an open top is made from 300 cm² of material. Find the largest possible volume.

Solution. Let x be the base side and h the height. Material: $x^2 + 4xh = 300$, so $h = \frac{300 - x^2}{4x}$. Then

$$V(x) = x^2h = \frac{300x - x^3}{4}, \quad V'(x) = \frac{300 - 3x^2}{4}.$$

$V' = 0$ at $x = 10$, and $V'' = -\frac{3x}{2} < 0$, a max. Then $h = 5$ and $V = 500$ cm³. The best box is twice as wide as it is tall. ■

- (e) A rectangular pen is split in half by a divider parallel to one side. The outer fence costs \$4 per meter and the divider \$2 per meter. With \$200, what is the largest possible area?

Solution. Let the pen be x by y meters with the divider of length y . Cost: $4(2x + 2y) + 2y = 8x + 10y = 200$, so $y = 20 - 0.8x$.

$$A(x) = x(20 - 0.8x) = 20x - 0.8x^2, \quad A'(x) = 20 - 1.6x.$$

$A' = 0$ at $x = 12.5$, and $A'' < 0$. Then $y = 10$, and the largest area is 125 m². ■

- (f) Find the points on the parabola $y = x^2$ closest to the point $(0, 2)$.

Solution. Minimize the squared distance (it has the same minimizer and no root).

$$D(x) = x^2 + (x^2 - 2)^2 = x^4 - 3x^2 + 4, \quad D'(x) = 4x^3 - 6x = 2x(2x^2 - 3).$$

Critical numbers $x = 0$ and $x = \pm \frac{\sqrt{6}}{2}$. $D(0) = 4$ while $D\left(\pm \frac{\sqrt{6}}{2}\right) = \frac{7}{4}$, so the closest points are $\left(\pm \frac{\sqrt{6}}{2}, \frac{3}{2}\right)$, at distance $\frac{\sqrt{7}}{2}$. The bottom of the parabola, $(0, 0)$, is a local max of the distance here, not the min. ■

19. Antiderivatives.

- (b) Find the most general antiderivative of $g(s) = \frac{4-2s+\sqrt{s^3}}{\sqrt{s}}$.

Solution. Split the fraction into powers of s first: $g(s) = 4s^{-1/2} - 2s^{1/2} + s$.

$$G(s) = 8s^{1/2} - \frac{4}{3}s^{3/2} + \frac{s^2}{2} + C.$$

- (g) If $f''(t) = 2 \cos t - 6 \sin t$, $f(0) = 1$, and $f'(0) = 4$, find $f(t)$.

Solution. $f'(t) = 2 \sin t + 6 \cos t + C_1$, and $f'(0) = 6 + C_1 = 4$ gives $C_1 = -2$.
 $f(t) = -2 \cos t + 6 \sin t - 2t + C_2$, and $f(0) = -2 + C_2 = 1$ gives $C_2 = 3$.

$$f(t) = -2 \cos t + 6 \sin t - 2t + 3.$$

- (i) A particle has $a(t) = 6t - 4$, $v(0) = 3$, $s(0) = 1$. Find $s(t)$.

Solution. $v(t) = 3t^2 - 4t + 3$ (the constant is $v(0)$), and $s(t) = t^3 - 2t^2 + 3t + 1$ (the constant is $s(0)$). ■

20. Riemann sums.

- (a) Use 6 rectangles with right endpoints to approximate the area under $\sin(x)$ on $[0, \pi]$.

Solution. $\Delta x = \frac{\pi}{6}$ and the right endpoints are $\frac{k\pi}{6}$ for $k = 1, \dots, 6$, with heights $\frac{1}{2}, \frac{\sqrt{3}}{2}, 1, \frac{\sqrt{3}}{2}, \frac{1}{2}, 0$.

$$R_6 = \frac{\pi}{6} (2 + \sqrt{3}) \approx 1.954.$$

The exact area is 2. ■

- (b) Approximate the area under $f(x) = x^2$ on $[0, 4]$ with $n = 4$ using right endpoints, left endpoints, and midpoints. Which are over and which are under?

Solution. $\Delta x = 1$.

$$R_4 = 1 + 4 + 9 + 16 = 30, \quad L_4 = 0 + 1 + 4 + 9 = 14, \quad M_4 = 0.25 + 2.25 + 6.25 + 12.25 = 21.$$

The exact area is $\frac{64}{3} \approx 21.33$. f is increasing, so the right endpoint is the tallest point of each strip and R_4 is an over estimate, while L_4 is an under estimate. M_4 comes out under here because the graph is concave up. ■

- (e) Write $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \sqrt{1 + \frac{3i}{n}}$ as a definite integral, then evaluate it.

Solution. Read $\Delta x = \frac{3}{n}$ and $x_i = 1 + \frac{3i}{n}$, so the interval starts at 1 and has length 3, and the height is $\sqrt{x_i}$.

$$\int_1^4 \sqrt{x} \, dx = \frac{2}{3} [x^{3/2}]_1^4 = \frac{2}{3}(8 - 1) = \frac{14}{3}.$$

21. The integral as area.

- (a) Evaluate $\int_0^2 (3x + \sqrt{4 - x^2}) \, dx$ using areas.

Solution. $\int_0^2 3x \, dx$ is a triangle with base 2 and height 6: area 6. $y = \sqrt{4 - x^2}$ is the top half of the circle of radius 2, so $\int_0^2 \sqrt{4 - x^2} \, dx$ is a quarter disk: area π .

Total: $6 + \pi$. ■

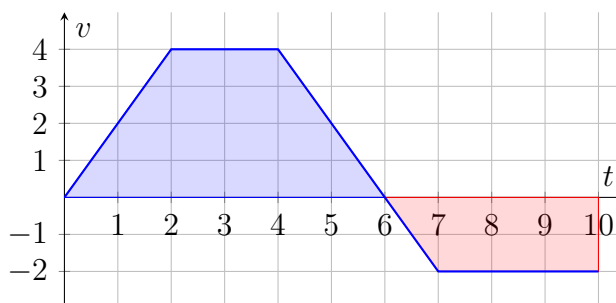
- (b) Evaluate $\int_{-4}^4 (2 - \sqrt{16 - x^2}) \, dx$ using areas.

Solution. $\int_{-4}^4 2 \, dx$ is a 8×2 rectangle, area 16. $\int_{-4}^4 \sqrt{16 - x^2} \, dx$ is a half disk of radius 4, area 8π . So the integral is $16 - 8\pi$, which is negative because most of the semicircle rises above the line $y = 2$. ■

- (d) Evaluate $\int_0^5 |2x - 4| \, dx$.

Solution. The graph is a V with its tip at $x = 2$. On $[0, 2]$ it is a triangle with base 2, height 4, area 4. On $[2, 5]$ it is a triangle with base 3, height 6, area 9. Total 13. ■

- (f) For the velocity graph below (v in m/s, t in s), find the displacement and total distance from $t = 0$ to $t = 10$, the position at $t = 10$ if $x(0) = 3$ m, the acceleration at $t = 5$, and when the particle is farthest from its start.



Solution. Displacement is the signed area under v . Above the axis (blue): triangle 4, rectangle 8, triangle 4, total 16. Below (red): triangle 1 and rectangle 6, total 7.

$$\text{displacement} = 16 - 7 = 9 \text{ m}, \quad \text{distance} = 16 + 7 = 23 \text{ m}.$$

The position at $t = 10$ is $3 + 9 = 12$ m. The acceleration at $t = 5$ is the slope of the segment from $(4, 4)$ to $(7, -2)$, which is -2 m/s^2 .

The particle moves forward while $v > 0$ and backward after $t = 6$, so it is farthest from the start at $t = 6$, 16 m away. ■

22. FTC, Part 1.

- (a) Calculate $f'(\pi/4)$ if $f(x) = \int_{-x}^x \cos(t) dt$, without finding an antiderivative.

Solution. Split at 0: $f(x) = \int_0^x \cos t dt - \int_0^{-x} \cos t dt$. By FTC 1 and the chain rule,

$$f'(x) = \cos x - \cos(-x) \cdot (-1) = 2 \cos x.$$

So $f'(\pi/4) = \sqrt{2}$. Both edges move outward as x grows, so the area gains $\cos$ at each end. ■

- (c) Find $g'(x)$ if $g(x) = \int_{x^2}^4 t\sqrt{1+t^2} dt$.

Solution. Flip the limits so the variable is on top: $g(x) = -\int_4^{x^2} t\sqrt{1+t^2} dt$. Then

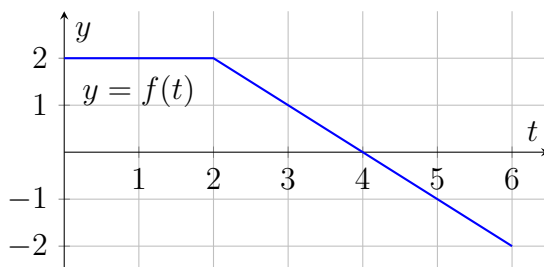
$$g'(x) = -x^2\sqrt{1+x^4} \cdot 2x = -2x^3\sqrt{1+x^4}.$$

- (e) Find $g'(x)$ if $g(x) = \int_{2x}^{x^3} \frac{1}{1+t^4} dt$.

Solution. Split at 0 and apply FTC 1 with the chain rule to each piece.

$$g'(x) = \frac{3x^2}{1+x^{12}} - \frac{2}{1+16x^4}.$$

- (f) Let $g(x) = \int_0^x f(t) dt$, where f is graphed below. Find $g(4)$, $g'(4)$, $g''(4)$, and where on $[0, 6]$ g is largest.



Solution. $g(4)$ is the area from 0 to 4: a 2×2 square plus a triangle with base 2 and height 2, so $g(4) = 4 + 2 = 6$.

By FTC 1, $g'(4) = f(4) = 0$, and $g''(4) = f'(4) = -1$, the slope of the line.

g increases while $f > 0$ and decreases once $f < 0$, so g is largest at $x = 4$, with $g(4) = 6$. ■

23. FTC, Part 2.

- (a) Calculate $\int_{32}^{34} 2x \, dx$ as an area, then with FTC 2.

Solution. The region under $y = 2x$ from 32 to 34 is a trapezoid with width 2 and heights 64 and 68: area $2 \cdot \frac{64+68}{2} = 132$.

With FTC 2, x^2 is an antiderivative: $34^2 - 32^2 = (34 - 32)(34 + 32) = 132$. ■

- (b) Evaluate $\int_1^4 \left(3\sqrt{x} - \frac{2}{x^2}\right) dx$.

Solution.

$$\int_1^4 \left(3x^{1/2} - 2x^{-2}\right) dx = \left[2x^{3/2} + \frac{2}{x}\right]_1^4 = \left(16 + \frac{1}{2}\right) - (2 + 2) = \frac{25}{2}.$$

- (f) What is wrong with $\int_{-1}^2 \frac{1}{x^2} dx = \left[-\frac{1}{x}\right]_{-1}^2 = -\frac{3}{2}$?

Solution. FTC 2 needs f continuous on $[a, b]$, and $\frac{1}{x^2}$ blows up at $x = 0$, inside $[-1, 2]$. The answer cannot be right anyway, because $\frac{1}{x^2} > 0$ and the integral of a positive function cannot be negative. (This integral is improper, and it diverges. Improper integrals come in Calculus 2.) ■

24. Substitution.

- (a) Evaluate $\int_1^2 x\sqrt{x-1} \, dx$.

Solution. Let $u = x - 1$, so $du = dx$ and $x = u + 1$. The bounds become $u = 0$ and $u = 1$.

$$\int_0^1 (u+1)u^{1/2} \, du = \left[\frac{2}{5}u^{5/2} + \frac{2}{3}u^{3/2}\right]_0^1 = \frac{2}{5} + \frac{2}{3} = \frac{16}{15}.$$

- (c) Show that u -substitution undoes the chain rule by computing $\int_a^b g'(h(x))h'(x) \, dx$.

Solution. Let $u = h(x)$, so $du = h'(x) \, dx$, and the bounds become $h(a)$ and $h(b)$.

$$\int_a^b g'(h(x))h'(x) \, dx = \int_{h(a)}^{h(b)} g'(u) \, du = g(h(b)) - g(h(a)).$$

By the chain rule, $g'(h(x))h'(x)$ is the derivative of $g(h(x))$, and FTC 2 gives the same answer directly. ■

(e) Evaluate $\int_0^1 x \cos\left(\frac{\pi x^2}{2}\right) dx$.

Solution. Let $u = \frac{\pi x^2}{2}$, so $du = \pi x dx$ and $x dx = \frac{du}{\pi}$. The bounds become 0 and $\frac{\pi}{2}$.

$$\frac{1}{\pi} \int_0^{\pi/2} \cos u du = \frac{1}{\pi} [\sin u]_0^{\pi/2} = \frac{1}{\pi}.$$

■

(i) Evaluate $\int_{-2}^1 x\sqrt{x+3} dx$.

Solution. Let $u = x + 3$, so $du = dx$, $x = u - 3$, and the bounds become 1 and 4.

$$\int_1^4 (u-3)u^{1/2} du = \left[\frac{2}{5}u^{5/2} - 2u^{3/2} \right]_1^4 = \left(\frac{64}{5} - 16 \right) - \left(\frac{2}{5} - 2 \right) = -\frac{8}{5}.$$

The answer is negative because $x\sqrt{x+3}$ is negative on most of $[-2, 1]$.

■

(k) Substitute $u = 2x - 1$ in $\int_{1/2}^1 (2x - 1) dx$ without evaluating. Approximate both the x -integral and the u -integral with 2 right-endpoint rectangles.

Solution. $du = 2 dx$, so $dx = \frac{du}{2}$, and $x = \frac{1}{2}, 1$ become $u = 0, 1$. The integral becomes $\int_0^1 \frac{u}{2} du$.

The x -integral: $\Delta x = \frac{1}{4}$, right endpoints $\frac{3}{4}, 1$, heights $\frac{1}{2}, 1$, sum $\frac{1}{4} \cdot \frac{3}{2} = \frac{3}{8}$. The

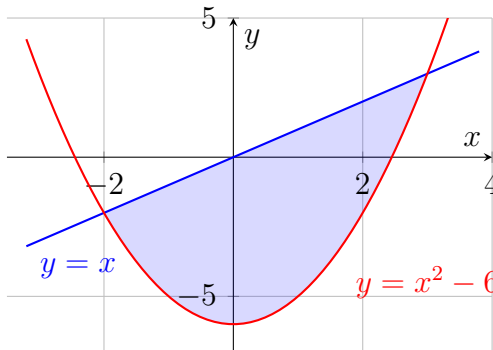
u -integral: $\Delta u = \frac{1}{2}$, right endpoints $\frac{1}{2}, 1$, heights $\frac{1}{4}, \frac{1}{2}$, sum $\frac{1}{2} \cdot \frac{3}{4} = \frac{3}{8}$.

They agree. The substitution makes every rectangle twice as wide and half as tall, so each area is unchanged. (Both integrals equal $\frac{1}{4}$.)

■

25. Area between curves.

(a) Sketch the region between $y = x$ and $y = x^2 - 6$ and find its area.



Solution. They meet where $x = x^2 - 6$, that is $(x - 3)(x + 2) = 0$: $x = -2, 3$. Between them the line is on top (at $x = 0$: $0 > -6$).

$$\int_{-2}^3 (x - x^2 + 6) dx = \left[\frac{x^2}{2} - \frac{x^3}{3} + 6x \right]_{-2}^3 = \frac{27}{2} + \frac{22}{3} = \frac{125}{6}.$$

■

- (c) For $y = x^2$ and $y = 8 - x^2$, set up the area integrating with respect to x , then y .

Solution. They meet at $x = \pm 2$, $y = 4$. In x : $\int_{-2}^2 ((8 - x^2) - x^2) dx = \frac{64}{3}$.

In y the right and left edges change formula at $y = 4$. For $0 \leq y \leq 4$ the width is $\sqrt{y} - (-\sqrt{y})$, and for $4 \leq y \leq 8$ it is $2\sqrt{8 - y}$.

$$\int_0^4 2\sqrt{y} dy + \int_4^8 2\sqrt{8 - y} dy = \frac{32}{3} + \frac{32}{3} = \frac{64}{3}.$$

■

- (d) Find the area between $x = y^2$ and $x = y + 2$.

Solution. They meet where $y^2 = y + 2$: $y = -1, 2$. The line is to the right (at $y = 0$: $2 > 0$).

$$\int_{-1}^2 (y + 2 - y^2) dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3} \right]_{-1}^2 = \frac{10}{3} + \frac{7}{6} = \frac{9}{2}.$$

In x this region would need two integrals, since the bottom edge changes formula at $x = 1$.

■

26. Average value.

- (a) Find the average value of $f(x) = \frac{1}{x^2}$ on $[1, 3]$.

Solution.

$$\frac{1}{3 - 1} \int_1^3 x^{-2} dx = \frac{1}{2} \left[-\frac{1}{x} \right]_1^3 = \frac{1}{2} \left(1 - \frac{1}{3} \right) = \frac{1}{3}.$$

■

- (c) Find the average value of $f(x) = x^2$ on $[0, 3]$ and a c in $[0, 3]$ with $f(c)$ equal to it.

Solution. $f_{\text{ave}} = \frac{1}{3} \int_0^3 x^2 dx = \frac{1}{3} \cdot 9 = 3$. Then $c^2 = 3$ gives $c = \sqrt{3}$, which is in $[0, 3]$. The Mean Value Theorem for integrals says such a c always exists for continuous f .

■

27. Volumes.

- (b) Find the volume when the region between $y = x$ and $y = x^2$ is revolved around the x -axis.

Solution. The curves meet at $x = 0, 1$, and $x \geq x^2$ there. A slice at x is a washer with $R = x$, $r = x^2$.

$$V = \pi \int_0^1 (x^2 - x^4) dx = \pi \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{2\pi}{15}.$$

■

- (e) Revolve the region between $y = 3x$ and $y = x^2$ around the y -axis, and then around the line $y = -1$.

Solution. They meet at $(0, 0)$ and $(3, 9)$. Around the y -axis, slice horizontally. At height y the region runs from the line $x = \frac{y}{3}$ out to the parabola $x = \sqrt{y}$, so $R = \sqrt{y}$ and $r = \frac{y}{3}$.

$$V = \pi \int_0^9 \left(y - \frac{y^2}{9} \right) dy = \pi \left(\frac{81}{2} - 27 \right) = \frac{27\pi}{2}.$$

Around $y = -1$, slice vertically. The radii are distances to $y = -1$: $R = 3x + 1$ and $r = x^2 + 1$.

$$V = \pi \int_0^3 ((3x + 1)^2 - (x^2 + 1)^2) dx = \pi \int_0^3 (-x^4 + 7x^2 + 6x) dx = \frac{207\pi}{5}.$$

■

28. Work.

- (a) If 2 J of work stretches a spring from 30 cm (natural length) to 42 cm, how much work stretches it from 35 cm to 40 cm? When pulled with 30 N, what is its length?

Solution. Work in meters: stretching 0.12 m takes $\int_0^{0.12} kx dx = 0.0072k = 2$, so $k = \frac{2500}{9}$ N/m.

$$W = \int_{0.05}^{0.10} kx dx = \frac{k}{2} (0.01 - 0.0025) = \frac{25}{24} \approx 1.04 \text{ J}.$$

With 30 N, $x = \frac{30}{k} = 0.108$ m, so the spring is $30 + 10.8 = 40.8$ cm long.

■

- (b) A force of 12 N holds a spring stretched from its natural length of 10 cm to 14 cm. How much work stretches it from 12 cm to 16 cm?

Solution. $12 = k(0.04)$ gives $k = 300$ N/m. The stretch goes from 0.02 m to 0.06 m.

$$W = \int_{0.02}^{0.06} 300x dx = 150 (0.0036 - 0.0004) = 0.48 \text{ J}.$$

■

- (c) A spring has natural length 10 cm and spring constant k . Compare the work W_1 from 10 cm to 15 cm with W_2 from 15 cm to 20 cm.

Solution.

$$W_1 = \int_0^{0.05} kx dx = \frac{k}{800}, \quad W_2 = \int_{0.05}^{0.10} kx dx = \frac{3k}{800}.$$

So $W_2 = 3W_1$. The force grows with the stretch, so the second 5 cm is pulled against a larger force. On the graph of $F = kx$, W_1 is a triangle and W_2 is a trapezoid three times its size.

■